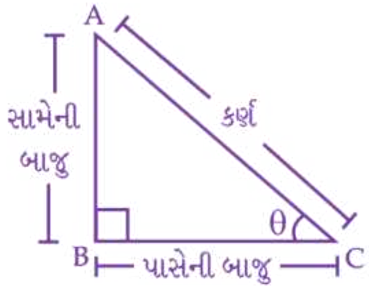
30. ત્રિકોણમિતિ

❑ સમજૂતી :



⇨ (સા. બા.) સામેની બાજુ = AB

⇨ (પા. બા.) પાસેની બાજુ = BC

⇨ કર્ણ = AC

⇨ $\frac{\text{સા.બા.}}{\text{કર્ણ}}$ $\frac{\text{કર્ણ}}{\text{સા.બા.}}$ (વ્યસ્ત કરતાં)

⇨ $\frac{\text{પા.બા.}}{\text{કર્ણ}}$ $\frac{\text{કર્ણ}}{\text{પા.બા.}}$ (વ્યસ્ત કરતાં)

⇨ $\frac{\text{સા.બા.}}{\text{પા.બા.}}$ $\frac{\text{પા.બા.}}{\text{સા.બા.}}$ (વ્યસ્ત કરતાં)

⇨ તો, $\frac{\text{સા.બા.}}{\text{કર્ણ}} = sin\theta$ $\frac{\text{કર્ણ}}{\text{સા.બા.}} = cosec\theta$

⇨ $\frac{\text{પા.બા.}}{\text{કર્ણ}} = cos\theta$ $\frac{\text{કર્ણ}}{\text{પા.બા.}} = sec\theta$

⇨ $\frac{\text{સા.બા.}}{\text{પા.બા.}} = tan\theta$ $\frac{\text{પા.બા.}}{\text{સા.બા.}} = cot\theta$

⇨ તો, આ બધાના વ્યસ્ત છે.

⇨ તેથી, $\sin\theta \times \text{cosec}\theta = 1$

$\cos\theta \times \sec\theta = 1$

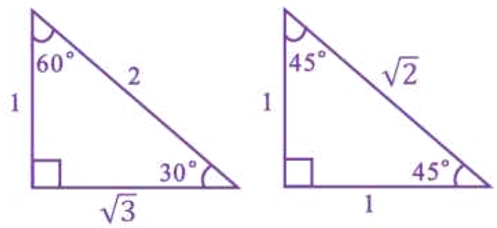
$\tan\theta \times \cot\theta = 1$

⇨ વ્યસ્ત,

$\sin\theta \to \text{cosec}\theta$

$\cos\theta \to \sec\theta$

$\tan\theta \to \cot\theta$

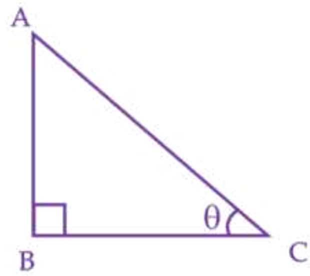


	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	અવ્યાખ્યાયિત
cosec	અવ્યાખ્યાયિત	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	અવ્યાખ્યાયિત
cot	અવ્યાખ્યાયિત	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

⇨ $\sin 0° = 0 \to \cos 0° = 1$

⇨ $\sin 90° = 1 \to \cos 90° = 0$

❖ પરિણામ :



(1) $AB^2 + BC^2 = AC^2$

⇨ $\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{\cancel{AC^2}}{\cancel{AC^2}}$

⇨ $\left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = 1$

⇨ $\left(\frac{\text{સા.બા.}}{\text{કર્ણ}}\right)^2 + \left(\frac{\text{પા.બા.}}{\text{કર્ણ}}\right)^2 = 1$

⇨ $\sin^2\theta + \cos^2\theta = 1$

(અથવા)

⇨ $\sin^2\theta = 1 - \cos^2\theta$

(અથવા)

⇨ $\cos^2\theta = 1 - \sin^2\theta$

(2) $AB^2 + BC^2 = AC^2$

⇨ $\frac{AB^2}{AB^2} + \frac{BC^2}{AB^2} = \frac{AC^2}{AB^2}$

⇨ $1 + \left(\frac{BC}{AB}\right)^2 = \left(\frac{AC}{AB}\right)^2$

⇨ $1+\left(\frac{\text{પા.બા.}}{\text{સા.બા.}}\right)^2=\left(\frac{\text{કર્ણ}}{\text{સા.બા.}}\right)^2$

⇨ $1+\cot^2\theta=\text{coses}^2\theta$

(અથવા)

⇨ $1=\text{cosec}^2\theta-\cot^2\theta$

(અથવા)

⇨ $\cot^2\theta=\text{coses}^2\theta-1$

(2) $AB^2+BC^2=AC^2$

⇨ $\frac{AB^2}{BC^2}+\frac{BC^2}{BC^2}=\frac{AC^2}{BC^2}$

⇨ $\left(\frac{AB}{BC}\right)^2+1=\left(\frac{AC}{BC}\right)^2$

⇨ $\left(\frac{\text{સા.બા.}}{\text{પા.બા.}}\right)^2+1=\left(\frac{\text{કર્ણ}}{\text{પા.બા.}}\right)^2$

⇨ $\tan^2\theta+1=\sec^2\theta$

(અથવા)

⇨ $\tan^2\theta=\sec^2A-1$

(અથવા)

⇨ $1=\sec^2\theta-\tan^2\theta$

⇨ $\tan^2A+1=\sec^2A$

(અથવા)

⇨ $\tan^2A=\sec^2A-1$

(અથવા)

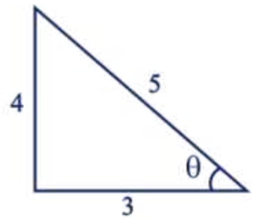
$1=\sec^2A-\tan^2A$

Triplet			
(1)	3	4	5
(2)	5	12	13
(3)	6	8	10
(4)	7	24	25
(5)	8	15	17
(6)	9	12	15
(7)	9	40	41
(8)	10	24	26
(9)	11	60	61
(10)	12	35	37
(11)	21	28	35
(12)	20	21	29

Type # 1

(1) જો $\sin\theta=\frac{4}{5}$ હોય તો $\cot\theta$ શોધો.

Solution :



⇨ Triplet : (3, 4, 5)

⇨ $\sin\theta=\frac{4}{5}$ (વ્યસ્ત કરતા ગુણોત્તર) $\rightarrow \text{cosec}\theta=\frac{5}{4}$

⇨ $\cos\theta=\frac{\text{પા.બા.}}{\text{કર્ણ}}\rightarrow\sec\theta=\frac{5}{3}$

$=\frac{3}{5}$

⇨ $\tan\theta=\frac{\text{સા.બા.}}{\text{પા.બા.}}\rightarrow\cot\theta=\frac{3}{4}$

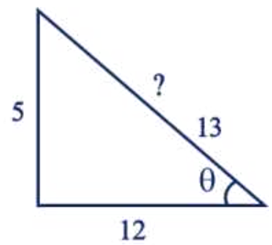
$\cot\theta=\frac{3}{4}$

(2) જો $\tan\theta=\frac{5}{12}$ હોય તો $\sin\theta+\cos\theta=$?

(A) $\frac{17}{13}$ (B) $\frac{12}{13}$

(C) $\frac{1}{2}$ (D) $\frac{5}{7}$

Solution :



⇨ Triplet : (5, 12, 13)

⇨ $\sin\theta+\cos\theta=\frac{5}{13}+\frac{12}{13}$

$=\frac{5+12}{13}$

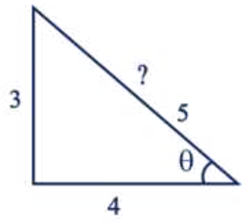
$=\frac{17}{13}$

$\boxed{\therefore \text{(A) } \frac{17}{13}}$

(3) **જો $\tan\theta = \frac{3}{4}$ હોય તો $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}$ = ?**

(A) −5 (B) −3 (C) −7 (D) −2

Solution :



⇨ Triplet : (3, 4, 5)

⇨ $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta} = \frac{\frac{3}{5}+\frac{4}{5}}{\frac{3}{5}-\frac{4}{5}}$

$= \frac{\frac{3+4}{\cancel{5}}}{\frac{3-4}{\cancel{5}}}$

$= \frac{7}{-1}$

$= -7$

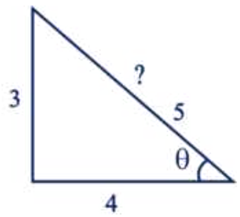
$\boxed{\therefore \text{(C) } -7}$

---

(4) **જો $\cot\theta = \frac{4}{3}$ હોય તો $2\sin^2\theta + 3\cos^2\theta$ =?**

(A) $\frac{1}{3}$ (B) $\frac{66}{25}$ (C) $\frac{40}{23}$ (D) $\frac{34}{25}$

Solution :



⇨ Triplet : (3, 4, 5)

⇨ $2\sin^2\theta + 3\cos^2\theta = 2\times\left(\frac{3}{5}\right)^2 + 3\left(\frac{4}{5}\right)^2$

$= 2\times\frac{9}{25} + 3\times\frac{16}{25}$

$= \frac{18}{25} + \frac{48}{25}$

$= \frac{18+48}{25}$

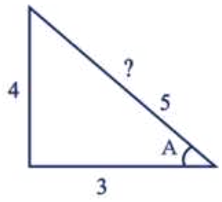
$= \frac{66}{25}$

$\boxed{\therefore \text{(B) } \frac{66}{25}}$

(5) **જો $\tan A = \frac{4}{3}$ હોય તો $\angle A$ ના cotA શોધો.**

(A) $\frac{3}{4}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Solution :



⇨ Triplet : (3, 4, 5)

⇨ $\sin A = \frac{4}{5}$ $\rightarrow$ $\operatorname{cosec} A = \frac{5}{4}$

⇨ $\cos A = \frac{3}{5}$ $\rightarrow$ $\sec A = \frac{5}{3}$

⇨ $\tan A = \frac{4}{3}$ $\rightarrow$ $\cot A = \frac{3}{4}$

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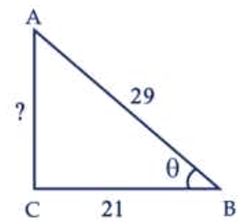
(6) **જેમા $\angle C$ કાટખૂણો હોય, તેવો કોઇ ΔACB લો. AB=29 એકમ, BC = 21 એકમ અને $\angle ABC = \theta$ હોય, તો (i) $\cos^2\theta + \sin^2\theta$, (ii) $\cos^2\theta - \sin^2\theta$ ની કિંમત શોધો.**

(A) 5, $\frac{11}{21}$ (B) $\frac{22}{23}$, 10 (C) 1, $\frac{41}{841}$ (D) $\frac{15}{17}$, 9

Solution :



⇨ Triplet : (20, 21, 29)

(i) $\cos^2\theta + \sin^2\theta$

⇨ $\left(\frac{21}{29}\right)^2 + \left(\frac{20}{29}\right)^2 = \frac{21^2+20^2}{29^2}$

$= \frac{441+400}{841}$

$= \frac{841}{841}$

$= 1$

(ii) $\cos^2\theta - \sin^2\theta$

⇨ $\left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = \frac{21^2-20^2}{29^2}$

$= \frac{441-400}{841}$

$= \frac{41}{841}$

$\boxed{\therefore \text{(C) } 1, \frac{41}{841}}$

(7) કાટકોણ ત્રિકોણ ABC માં ખૂણો B કાટખૂણો છે. જો $\tan A = 1$ હોય તો $2 \sin A \cdot \cos A = ?$

(A) 0 (B) 1

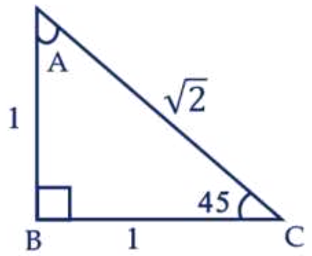
(C) 2 (D) $\sqrt{2}$

Solution :

⇨ $\tan A = 1$

⇨ $\tan A = \frac{1}{1}$

⇨ $\tan A = \frac{1}{1}$ $A = 45^\circ$



⇨ $2 \sin \cdot \cos A = 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{\cancel{2}}{\cancel{2}} = 1$

$\boxed{\therefore \text{(B) } 1}$

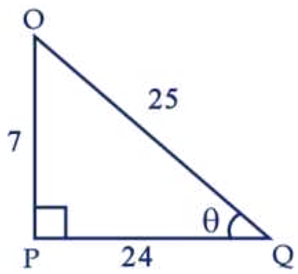
---

(8) ΔOPQ માં, P કાટખૂણો છે, OP = 7 સેમી અને OQ − PQ = 1 સેમી હોય તો cos Q નું મૂલ્ય શોધો.

(A) $\frac{12}{13}$ (B) $\frac{24}{25}$

(C) $\frac{11}{25}$ (D) $\frac{19}{21}$

Solution :



⇨ Triplet : (7, 24, 25)

⇨ OQ − PQ = 1

⇨ 25 − 24 = 1

⇨ $\cos\theta = \frac{24}{25}$

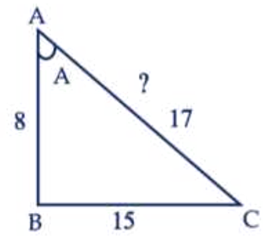
$\boxed{\therefore \text{(B) } \frac{24}{25}}$

(9) જો $15 \cot A = 8$ હોય, તો sec A શોધો.

(A) $\frac{17}{8}$ (B) $\frac{24}{25}$

(C) $\frac{11}{25}$ (D) $\frac{19}{21}$

Solution :



⇨ Triplet : (8, 15, 17)

⇨ $15 \cot A = 8$

⇨ તો, $\cot A = \frac{8}{15}$

⇨ $\sec A = \frac{17}{8}$

$\boxed{\therefore \text{(A) } \frac{17}{8}}$

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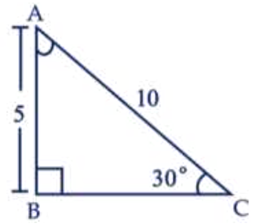
(10) ΔABC માં B કાટખૂણો છે, AB = 5 સેમી અને $\angle ACB = 30^\circ$ હોય તો બાજુ BCની લંબાઈ શોધો.

(A) $7\sqrt{4}$ (B) $5\sqrt{3}$

(C) $1\sqrt{3}$ (D) $9\sqrt{11}$

Solution :



⇨ $\text{Sin } 30^\circ = \frac{AB}{AC}$

$\frac{1}{2} = \frac{5}{AC}$

$AC = 2 \times 5$

AC = 10 સેમી

⇨ AC = 10

⇨ $BC^2 = AC^2 - AB^2$

⇨ $BC^2 = 100 - 25$

⇨ $BC^2 = 75$

⇨ $BC^2 = 25 \times 3$

⇨ $BC = 5\sqrt{3}$ સેમી

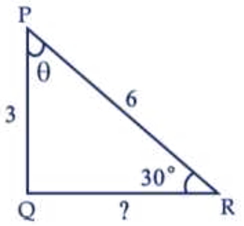
$\boxed{\therefore \text{(B) } 5\sqrt{3} \text{ સેમી}}$

(11) ΔPQR માં Q કાટખૂણો છે, PQ = 3 સેમી અને PR = 6 સેમી હોય, તો $\angle$QPR અને $\angle$PRQ શોધો.

(A) $80°, 40°$ (B) $25°, 90°$

(C) $60°, 30°$ (D) $45°, 60°$

Solution :



⇨ $\text{Sin R} = \frac{\cancel{3}^{\,1}}{\cancel{6}_{\,2}}$

⇨ $\angle PRQ = R = 30°$

⇨ R ની સામેનો ખૂણો P છે.

⇨ $\angle QPR = 2 \times R = 2 \times 30 = 60°$

⇨ $\angle PRQ = 30°$

⇨ $\angle$R એ $30°$ નો છે. તેની સામે ખૂણો બમણો હોય.

⇨ $\angle$P એ $60°$ નો થાય.

$\boxed{\therefore \text{ (C) } 60°, 30°}$

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(12) cos A = 4 sin A તો, tan A = ________

(A) $\frac{1}{4}$ (B) $\frac{5}{4}$ (C) 5 (D) $\frac{4}{5}$

Solution :

⇨ cos A = 4 sin A

⇨ $\frac{1}{4} = \frac{\sin A}{\cos A}$

⇨ $\tan A = \frac{1}{4}$

$\boxed{\therefore \text{ (A) } \frac{1}{4}}$

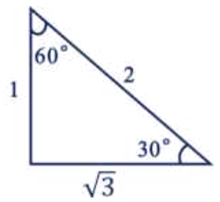
---

(13) $\frac{2\tan 30^0}{1+\tan^2 30^0}$ = ____

(A) $\sin 60°$ (B) $\cos 60°$

(C) $\tan 60°$ (D) $\sin 30°$

Solution :



⇨ $$\frac{2\tan 30°}{1+\tan^2 30°} = \frac{2 \times \frac{1}{\sqrt{3}}}{1+\left(\frac{1}{\sqrt{3}}\right)^2}$$

$$= \frac{\frac{2}{\sqrt{3}}}{1+\frac{1}{3}}$$

$$= \frac{\frac{2}{\sqrt{3}}}{\frac{3+1}{3}}$$

$$= \frac{\frac{2}{\sqrt{3}}}{\frac{4}{3}}$$

$$= \frac{\cancel{2} \times 3}{\cancel{4}_{\,2} \times \sqrt{3}}$$

$$= \frac{\sqrt{3} \times \cancel{\sqrt{3}}}{2 \times \cancel{\sqrt{3}}}$$

$$= \frac{\sqrt{3}}{2}$$

⇨ તો, $\sin 60°$

$\boxed{\therefore \text{ (A) } \sin 60°}$

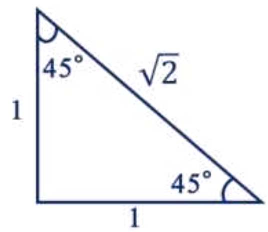
---

(14) $\frac{1-\tan^2 45^0}{1+\tan^2 45^0}$ = ____

(A) $\tan 90°$ (B) 1

(C) $\sin 45°$ (D) 0

Solution :



⇨ $$\frac{1-\tan^2 45°}{1+\tan^2 45°} = \frac{1-(1)^2}{1+(1)^2}$$

$$= \frac{1-1}{1+1}$$

$$= \frac{0}{2}$$

= 0 (શૂન્ય)

$\boxed{\therefore \text{ (D) } 0}$

(15) જ્યારે A°= ____ હોય, ત્યારે sin 2A = 2 sin A સત્ય હોય.

(A) 0° (B) 30° (C) 45° (D) 60°

Solution :

⇨ $\sin(2 \times 0) = 2\sin 0^\circ$

⇨ $\sin 0^\circ = 2 \times \sin 0^\circ$

$0 = 2 \times 0$

$0 = 0$

તો, $A = 0$

$\boxed{\therefore \text{(A) } 0^\circ}$

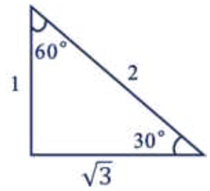
---

(16) $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ} =$ ____

(A) $\cos 60^\circ$ (B) $\sin 60^\circ$

(C) $\tan 60^\circ$ (D) $\sin 30^\circ$

Solution :



⇨ $\dfrac{2\tan 30^\circ}{1-\tan^2 30^\circ} = \dfrac{2\times\dfrac{1}{\sqrt{3}}}{1-\left(\dfrac{1}{\sqrt{3}}\right)^2}$

$= \dfrac{\dfrac{2}{\sqrt{3}}}{1-\dfrac{1}{3}}$

$= \dfrac{\dfrac{2}{\sqrt{3}}}{\dfrac{3-1}{3}}$

$= \dfrac{\dfrac{\cancel{2}}{\sqrt{3}}}{\dfrac{\cancel{2}}{3}}$

$= \dfrac{3}{\sqrt{3}}$

$= \dfrac{\sqrt{3}\times\sqrt{3}}{\sqrt{3}}$

$= \sqrt{3}$

$= \tan 60^\circ$

$\boxed{\therefore \text{(C) } \tan 60^\circ}$

(17) $\cos^2 A + \sin^2 A = 1$ તો $\cos^2 A = ?$

(A) $2-\sin^2 A$ (B) $4-\sin^2 A$

(C) $1-\sin^2 A$ (D) $5-\sin^2 A$

Solution :

⇨ $\sin^2 A + \cos^2 A = 1$

(અથવા)

⇨ $\sin^2 A = 1 - \cos^2 A$

(અથવા)

⇨ $\cos^2 A = 1 - \sin^2 A$

$\boxed{\therefore \text{(C) } 1-\sin^2 A}$

---

(18) $\operatorname{cosec}^2 A - \cot^2 A = ?$

(A) 1 (B) 2 (C) 3 (D) 4

Solution :

⇨ $\operatorname{cosec}^2 A - \cot^2 A = 1$

(અથવા)

⇨ $\operatorname{cosec}^2 A = 1 + \cot^2 A$

⇨ $\operatorname{cosec}^2 A - 1 = \cot^2 A$

$\boxed{\therefore \text{(A) } 1}$

---

(19) $\sec^2 A - \tan^2 A = ?$

(A) 1 (B) 2 (C) 3 (D) 4

Solution :

⇨ $\sec^2 A - \tan^2 A = 1$

(અથવા)

⇨ $\sec^2 A = 1 + \tan^2 A$

⇨ $\tan^2 A = \sec^2 A - 1$

$\boxed{\therefore \text{(A) } 1}$

---

(20) $\dfrac{\sin 18^\circ}{\cos 72^\circ} = ?$

(A) 1 (B) 2 (C) 3 (D) 4

Solution :

⇨ **કોટિકોણ :**

$\cos A = \sin(90 - A)$

$\sec A = \operatorname{cosec}(90 - A)$

$\tan A = \cot(90 - A)$

⇨ **રીત : 1**

⇨ $\sin 18^\circ = \cos(90 - 18^\circ)$

$= \cos 72^\circ$

$= \dfrac{\cos 72^\circ}{\cos 72^\circ}$

$= \dfrac{1}{1}$

$= 1$